# A Semantic Comparison of Legacy and Modern Blockchain Interoperability Protocols Using Process Algebra

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# Chapter 1

# Introduction

The blockchain technology first gained its popularity as a decentralised ledger focusing on the exchange of asset or cryptocurrency like Bitcoin . Initially, this design provides solution to the problem of double-spending in digital money without the need of third-party entity like banks. Every transaction in blockchain is publicly recorded in a way that they cannot be changed (tamper-proof) using cryptography. This ensures trust between the participants in the network eventhough they do not know each other personally.

Today, blockchains have evolved from simply exchanging currency to becoming a general-purpose platform with the introduction of Ethereum and smart contracts. For this reason, it became programmable that eventually led its use in different sectors like in supply chain, healthcare, voting, and Internet of Things (IoT). At this point, blockchain is no longer about financial transaction; it has become a means of safe storage and sharing of information that depends on the needs of individual industries.

This growth in its capabilities led to an array of independent blockchain platforms that together contributed to the problem of interoperability. Despite their growth both in their capabilities and number, blockchains have not yet been widely adopted or succesfully integrated into today's mainstream technological systems. To address this, they need not only solutions to the issues of decentralisation, scalability, and security (the blockchain trilemma), but also to their capability to communicate with each other. Without interoperability, they will continue to remain only as information silos or disconnected islands of information that cannot exchange value and data directly.

This limits the real potential of blockchain technology.

Solutions to interoperability exist like cross-chain bridges, relays, and sidechains. In recent literature, blockchain interoperability (BI) is divided into two aspects: digital asset interoperability (for cryptocurrency) and arbitrary data interoperability (for general information). This difference has been used by recent papers in order to understand the different logic and mechanisms needed by each category. Because of this, the issue of semantic interoperability surfaces. It is the challenge on how the meaning and correct interpretation of data remain consistent while they are being transferred to different blockchain networks.

## 1.1 Project Context

The evolution of the solutions for BI can be divided into three generations:

- First Generation (Asset-Only): These are protocols focusing only to the exchange of currencies. Hashed Time-Lock Contracts (HTLC) and atomic swaps are examples of this.
- Second Generation (Data-Only): Systems like oracle relays and simple message passing are designed simply for delivering information or state changes between blockchains.
- Third Generation (Unified Protocols): Modern frameworks like IBC (Cosmos), Polkadot XCM, and Axelar claims to be capable of handling both assets and data in using one unified structure.

While the legacy protocols (first and second generations) are specifically designed for single purpose, the recent unified protocols posesses both capabilities. Nonetheless, it is unclear whether they truly combine the behaviours of assets and data, or they only stack the semantics of assets on top of the transportation of data. Currently, there are no formal semantic comparison yet that uses mathematical processes to analyse this difference.

This research is important for blockchain developers, bridge designers, formal verification engineers, and interoperability researchers. This research will use protocol specifications in order to come up with Labelled Transition Systems (LTS) and traces. Using bisimulation equivalence, this research will analyse if unified protocols truly replaces legacy designs or if the semantics of assets and data are so different that they cannot be unified.

## 1.2 Purpose and Description

The main purpose of this research is to make a formal semantic comparison between the observable behaviours of the legacy asset-only protocols, legacy data-only protocols, and modern unified protocols using process algebra and LTS.

This research will give formal specifications for three representative protocols (one for each generation) using mathematical technique of CSP process algebra. From these specifications, semantic models will be extracted in the form of LTS in order to compare their traces and branching structure using trace equivalence and branching bisimulation.

The importance of this research can be seen from the following stakeholders:

- Blockchain Developers: To have a guideline whether to use unified protocols or stay with the legacy designs.
- Bridge Auditors: To have deeper understanding in the difference of behaviours that can lead to security issues.
- Formal Methods Researchers: To give a single framework for systemic comparison on interoperability protocols.
- Industry Practitioners: To provide stonger evidence whether the unified claims of new protocols are truly valid in the semantic aspect.

In order to guide this research, the following questions are posed:

1. How can asset-only, data-only, and unified BI protocols be formally specified using process algebra?
2. What are the observable behaviours (traces, branching structures, deadlocks) that will surface from the semantic models of each category?
3. Does the unified protocol show behaviours bisimilar to the composition of asset-only and data-only behaviours, or do they show behavioural conflicts that cannot be unified?

## 1.3 Objectives

The main objective of this research is to formally specify and semantically compare the observable behaviours of legacy asset-only, legacy data-only, and modern unified BI protocols using LTS and process algebra. In order to achieve this, the following specific objectives must be accomplished:

- Chose three representative protocols, one for each category.
- Model each protocol using process algebra while identifying their atomic actions, alphabets, and composition operators.
- Extract LTS models for each protocol.
- Identify observable traces and execution paths from each LTS model to analyse the flow of operations.
- Compare the three models using trace equivalence and branching simulation criteria.
- Idenftify if the unified protocol can simulate the behaviours of both legacy protocols or if there exist behavioural conflicts between them.

## 1.4 Scope and Limitations

### 1.4.1 Scope

This research focuses on three particular protocol: HTLC for asset-only, Oracle Relay for data-only, and IBC/XCM for unified category. Only core interaction patterns will be modelled in this research. The behavioural comparison will only use trace equivalence and branching bisimulation as the main criteria.

### 1.4.2 Limitations

This research does not include modelling of economic incentives like gas costs, slashing mechanisms, or liquidity contraints. This will not also analyse security properties like double-spending resistance or fraud proofs beyond what will be shown from the obersvable behaviours. Since only three protocols will be analysed, the results can be insuffiecient to generalise to all kinds of asset-only, data-only, or unified protocols. The formal models will abstract or remove implementation details of smart contracts. This research does not use empirical validation or implementation of the actual protocol instance.

# Chapter 2

# Review of Related Literature

In the current landscape of blockchain technology, the emergence of heterogeneous networks has revealed the term "information isolated isalands" or "value silos" [109] [19] [119]. Although individual blockchains are STRONG/GREAT as independent ledgers, the lack of communication between them limits their potential widespread adoption [46] [31] [68]. Blockchain interoperability then is not just a feature; but a critical interaction between distributed systems [109] [62] [68].

It is vital to understand that every blockchain acts as an independent state machine with its own concensus rules and encryption standards [19] [15]. Since there are no shared state or global clock between these different blockchains, their interactions must happen asynchronously; they must instead use message passing and state synchronisation [109] [113]. In other words, blockchain interoperability is a process wherein one source chain is trying to change the state of a target chain by means of delivering information or assets under certain securoty assumptions [109] [110] [16].

Most of study in the literature focus on the blockchain interoperability architecture. That is, whether it uses relays, sidechains, or notaries [54] [24] [61] [29]. Nevertheless, recent studies have put emphasis that architecture only gives stucture and real security and effectiveness of interoperability depends on the behaviour of the system [66] [65] [68]. This behaviour dictates how systems guarantee ACID (atomicity, consistency, isolation, durability) properties in an asynchronous network [109] [19] [110] [62].

This means that the sequencing of state transitions, timing of cyrptographic proofs, and reactions to failure or timeout reveals the system's behaviour [83] [32] [36] [113].The analysis of behaviours,

instead of the static design, gives deeper understanding on how different components (such as smart contracts and relayers) interact during different processe. That is so to achieve semantic interoperability where the meaning and context of data or asset from the source chain gets accurately interpreted and valid to the target chain [13] [98] [62].

## 2.1 Approaches to Blockchain Interoperability

Surveying the literature, it is noticeable that the classifications of interoperability reflects the underlying logic on how to trust and treat information between blockchains [109] [85] [111]. Although categorical labels like notary schemes, sidechains/relays, and hash-locking are usually used, there is a deeper abstraction that surfaces in recent literature. That is, the perspective of looking at blockchain interoperability as a multi-layered problem derived from the OSI model of the internet [46] [36].

The common abstraction from most of the sources is the **source-target model**. In this model, an operation from the source chain needs to trigger a corresponding state change to the target chain [36] [31]. Nonetheless, here comes the critical difference on how this interaction gets seen:

**Interoperability as Transfer:** In this perspective, the logic focuses on migration or transfer of ownership. Every transaction must follow all-or-nothing atomicity in order to avoid double-spending [109] [64] [36]. Here belongs asset-based protocols that ensure that the value will not vanish or duplicated between silos [13] [111].

**Interoperability as Communication:** Here, the system gets treated as a **message-oriented service** [109] [54] [48]. The goal is not to transfer the content of the source chain but to invoke a functionality in the target chain using arbitrary data [85] [15]. This is the logic of the data-based protocols like Arbitrary Message Passing (AMP) [10].

It is critical to make the distinction between asset-based and data-based interoperability because they usually remain implicit in a number of older literature. For instance, in the classical SoK (Systematisation of Knowledge) of Buterin, the focus was always in value exchange or value islands [109] [2] [85]. However, in the newer frameworks, there is an explicit division in the domain between **Asset Interoperability** (migration/exchange) and **Data Interoperability** (cross-chain calls/integration) [36] [31] [62]. This gap is important as a number of current models are designed for assets but gets applied to data or vice-versa without taking each different semantic requirements

into consideration [13] [62].

This differeng individual trust boundary and security assumptions of each approaches contributes to the fragmented landscape of blockchain interoperability[13]. Although there is an attemot to have a Generic Interoperability Paradigm comprising of Setup, Commit, Verify, and Abort phases, the literature has not fully explained how these phases differ in their **observable behaviour** when assets are passed isnstead of arbitrary data [36] [62]. This forms the need of a **comparative semantic analysis** to see if these models are enough to encompass the behaviour of both asset and data exchanges [62].

## 2.2 Asset-Based Interoperability

The literature treats asset-based interoperability not solely as a tool to transfer currency but also as a **closed system of value synchronisation** [85] [31]. The fundamental logic of these protocols is the preservation of the INTEGRITY of assets while they are moving between different ledgers [64] [85] [13].

### 2.2.1 System Perspective

As a system, asset-based interoperability comprises the following interdependent components: **Chains (Source and Target):** These serve as state machines that record ownership [109] [36] [31]. **Smart Contracts (Custodian vs. Debt Issuer):** In the source chain, there is usually a custodian contract (vault) that locks asset; while in the target chain, there is a debt issuer (money printer) that makes synthetic representation or wrapped tokens [81] [99]. **Users and Intermediaries:** Included here is the initiator (user), relayers that transfer proofs, and attesters that verify if the lock event is valid before doing the minting [81] [66] [13].

This shows that asset-based interoperability is a content-aware system. Meaning, it is necessary that the protocol knows the specific semantics of asset (e.g. ERC-20 standard) in order to ensure that the minted token is semantically equal to the original asset [66] [85] [36].

### 2.2.2 Behavioural Patterns: Logic of Ordering

The observable behaviour of asset-based protocols follow a strict sequential execution flow. Based on existing formal models (such as Event-B analysis), the usual sequence follows: **Lock/Burn**

**(Source):** Immobilisation of the original value [81] [66] [36]. **Verification:** Passing the Merkle proof or SPV (Simplified Payment Verification) proceeding to the target chain [109] [69] [19] [36]. **Mint/Claim:** Producing of an equal value at the target chain [66] [97] [62].

This shows that there is happens-before relationship. The mint transaction at the target chain has causal dependency on the lock transaction at the source chain [66] . If this sequence is not followed, there is a risk of atomicity failure where there is a possibility of minting assets that are not locked; resulting to double-spending [64] [13] [36] [61].

### 2.2.3 Invariants and Safety Guarantees

The major invariant that safeguards this system is the Global Supply Invariant. This refers to maintaning a constant value to the overall count of asset units in all networks [85] [61] [19].

It is critical that the safety (no bad behaviour or duplication) of asset-based protocols gets prioritised over the liveness (fast termination) [66] [19]. Because of this, the observable behaviour has long confirmation periods (to wait the finality from the source chain) and competition periods (for fraud proofs) [36] [109] [110] [71]. These delays are not just performance issue but a behavioural necessity in order to protect the atomicity of the system [32] [69] [31].

Nevertheless, this behaviour results to capital inefficiency [99] [77] [62]. That is because of the locking of assets, the liquidity gets trapped in the contracts, that serves as a big limitation that is not usually seen in data-based approaches [99] [36] [10]. This creates the gap in the literature where even though the safety of asset transfers can be formally modelled, there is insufficient comparison if this strict ordering is also needed for arbitrary data exchange [66] [65].

## 2.3 Data-Based Interoperability

Unlike asset-based interoperability where it is illustrated as a mechanism of changing ownership, the literature describes data-based interoperability as a decentralised communication system [69] [19] [54]. If asset-based focuses on liquidity, data-based on the other hand focuses on the arbitrary message passimg (AMP). This is a mechanism where the system acts as a transport layer for whatever type of information [35] [10].

### 2.3.1 System Perspective: The TCP/IP of Blockchains

Being a communication system, its architecture comprises of the following interdependent components: **Chains and Modules:** Every blockchain hosts modules or handlers (smart contracts). They serve as entry and exit points of data [35] [32]. **Relayers and Connectors:** Since blockchains are isolated islands and incapable of sending data natively, they need off-chain relayers (such as Hermes) that scans blocks for pending messages and transfer them to the target chain [35] [32]. **Channels and Connections:** These are the stateful abstractions that maintains logical link between two chains that gives ordering and authentication to packets [19] [113].

The literature sees the system as a payload-agnostic service. This shows that the protocols do not care whether what is inside the packets is a token transfer or a complex smart contract call. Its purpose is to ensure that the opaque bytes are delivered with integrity [54] [35] [32] [19].

### 2.3.2 Behavioural Patterns: Handshake and Asynchrony

The observable behaviour of data-based interoperability is guided by handshake protocols and asynchronous execution [69] [19] [114]. **Handshake Process:** This is unlike the simple lock-and-mint; data-based systems follows a strict 4-step handshake (OpenInit -> OpenTry -> OpenAck -> OpenConfirm) in order to establish a connection [19] [114]. This behaviour is needed to ensure that both chains are ready to accept data under the same security assumptions [114]. **Asynchronous Message Passing:** The interaction here is non-blocking. The source chain delivers a message and can proceed to other operations while the packet is in-flight [69] [19]. The success of the interaction depends on Acknowledgements (ACK) or Timeouts. They are observable events that dictates if an operation is deemed Delivered or Expired [32] [114].

### 2.3.3 Comparison to Asset-Based Interoperability

The fundamental difference between the two is rooted on their interaction logic: **State-Dependent vs. Instruction-Dependent:** Unlike the asset-based that depends on state provenance (it needs the verification of the lock state before minting), data-based behaviour is instruction-dependent [112] [36]. Its aim is to trigger a functionality in the target system that will work under its own rules [109] [15]. **Atomicity vs. Eventual Delivery:** In asset transfers, atomicity is commonly all-or-nothing under a capital constraint [85] [13]. Meanwhile, in data exchange, it prioritises liveness and delivery

guarantees. It means that every message should have a definite result (accepted or timeout) inside the asynchronous network [19] [85] [114].

The behaviour here depends on the reliabilit of the transport layer. Nonetheless, the critique in the current literature is the dependency of these systems to untrusted layers [114] [35]. Formal analyses (such as the TLA+) shows that even the standardized protocols can have trapped packets or incomplete handshakes if the relayers' behaviour cannot be properly modelled [113]. This pushed the need for deeper comparative semantic analysis to see how it is possible to converge the safety of assets and the liveness of data in one behavioural model.

## 2.4 Formal Methods to Blockchain Interoperability

In the study of interoperability protocols, simple architecture diagrams alone are insufficient to guarantee the security and correctness of systems. Formal Methods surfaces in the literature as critical tools to avoid design flaws that can lead to large financial losses [113] [65]. In surverying the literature, it is noticeable that each formal method has its own modelling philosophy that dictates what aspects of behaviour are being emphasized.

### 2.4.1 Modelling Philosophies and Methodogies

From what has been gathered in the literature, formalisations done on blockchain interoperability is divided into four major philosophies: **Event-B (State + Invariants):** This approach in focused on modelling the system as a discrete transition system [65]. The strength of this is in verifying the safety properties of the system using mathematical proofs and invariants. For example, this is used to ensure that a token cannot be minted if it is not yet officially locked on the source chain [66]. Nevertheless, this philosophy has its own limitation in capturing the liveness (or the guarantee that something will happen at the end), and is usually struggles in modelling asynchronous timing issues [66] [65]. **TLA+ (Traces + Temporal Behaviour):** In data-based protocols like IBC, TLA+ is the dominating method [114] [66]. Its philosophy is rooted in Linear Temporal Logic where the behaviour is seen as a sequence of states or traces. This is effective in detecting bugs in asynchronous communication like trapped packets or incomplete handshakes [114]. However, because of its high level of abstraction, it usually disregards the internal logic of smart contracts [65] [114]. **Abstract State Machine or ASM (Abstrac Execution):** The philosophy is to

abstract the whole blockchain as a deterministic transition function [69]. Its focus is the operational semantics of the remote calls and how they maintain atomicity in multi-chain transactions. This focuses mmore on the execution flow more than the static invariants of assets. **CSP# and pi-calculus (Interaction + Concurrency):** These methods focus on process algebra [108]. Its philosophy is in the treatment of interoperability as a series of inter-process communications. This is useful in modelling interactive protocols like atomic swaps where it is needed that two untrusted parties simultaneously do an action.

## 2.5 Synthesis and Research Gap

In surveying the literature, a clear pattern surfaces: although the improvement in the technical solutions to blokchain interoperability is growing fast, its theoretical and formal foundation still remains slow and fragmented. The following discussions lay outs the critical gaps that reasons the need for this research.

### 2.5.1 Fragmentaion in Knowledge and Technology

Every study in the current literature focuses on particular silo or isolated implementations. It can be observed that the current research landscape is divided into walled gardens where one group of the literature is exclusively discussing asset-specific bridges while the others solely focus on messaging-agnostic hubs. For this reason, the understanding of interoperability is not being hollistic. This fragmentation is not only technical but also conceptual; there is no unity in the literature on what really is the standard interaction process to be followed by all protocols.

### 2.5.2 Lack of Unified Abstraction

There is no unified way to represent the semantics of cross-chain interactions using a framework that is not dependent on codes. Most of the gathered studies give ad hoc architectures or intentionally made for one particular use case. This shows that the literature currently has insufficient behavioural primitives. These are guiding action units like locking, proving, or emitting that can be used to describe any protocol whether it is an asset-based or data-based. Without this abstraction, developers and researchers will struggle in verifying the security of systems when different protocols gets interconnected.

### 2.5.3 Lack in Direct Comparison of Behaviours

There is direct comparison done on the behaviour of asset-based and data-based protocols inside one formal semantic framework. Although there are surveys that list features such as whether they are decentralised or have notary, they do not survey the fundamental difference in terms of interaction logic. Asset-based protocols are described as state-dependent (it needs lock before minting), while data-based are instruction-depended (triggering functions) but the implication of this difference in their observable behaviour is not fomally written. Meaning, although their goal is known, their difference in their execution traces cannot be formally shown.

### 2.5.4 Methodological Gap in Formal Modelling

Most of the current formal models are state-based, not interaction-based. The literature is currently limited to one dichotomy: The usage of Event-B for verifying static invariants and safety of assets and the usage of TLA+ to survey liveness and traces of data packets. This gap is having no model that can unify the low-level state transitions (ownership changes) and high-level messaging flow (function calls) in one language. This does not directly analyses the interaction of value and data as connected manifestation of system behaviour.

### 2.5.5 Conclusion and Research Direction

In conclusion, there is a need to have an approach that will abstract these protocols as formal systems of interaction instead solely as implementations. The current gap in the literature is not just only a topic of fastness or throughput, but the lack of deep semantic understanding on how blockchain interoperability operates on the behavioural level. One possible way is the usage of a comparative semantic analysis that focus on the creation of formal model for asset and data transfers. This research aims to fill the aforementioned gaps by means of identifying the observable behaviours that will give clarity to the similarities and differences of the two paradigm. This will serve as a guideline for making more robust and standardized interoperability frameworks in the future.

Table 2.1: Matrix

| Citation | Type | Interoperability Type | Formalisation | Limitations |
|---|---|---|---|---|
| [99] | Survey/Protocol Design | Asset | | |

| | | | | |
|---|---|---|---|---|
| [81] | Survey/SoK | Hybrid (bridges) | - | Lacks formal system model |
| [24] | | | | |
| [10] | | | | |
| [62] | | | | |
| [35] | | | | |
| [51] | | | | |
| [36] | | | | |
| [2] | | | | |
| [46] | | | | |
| [55] | | | | |
| [66] | | | | |
| [19] | | | | |
| [69] | | | | |
| [64] | | | | |
| [65] | | | | |
| [31] | | | | |
| [98] | | | | |
| [77] | | | | |
| [60] | | | | |
| [108] | | | | |
| [114] | | | | |
| [9] | | | | |
| [29] | | | | |
| [61] | | | | |
| [111] | | | | |
| [13] | | | | |
| [119] | | | | |
| [45] | | | | |
| [54] | | | | |
| [32] | | | | |

[71]

[112]

[97]

[48]

[83]

[85]

[68]

[15]

[109]

# Chapter 3

# Theoretical Framework

## 3.1 Systems and System Behaviour

A **system** is formally composed of two major aspects: process and data. In this framework, processes serve as control mechanisms that runs and manipulates data. Meanwhile, data remains static and passively comprises the system [39]. The composition of these elements formsa structure that can perform operations according to specified rules [87].

The **behaviour** of a system refers to the sum of its ability to interact or communicate. A behaviour is **observable** because it can be seen from its outside operations; particularly, the interaction of the inputs it accepts and the outputs it produces [76]. Every elementary step or operation that happens in the system is called **transition** that changes the system from its current configuration to another state [87].

A **Blockchain** is an example of a system that can be treated as a transaction-based **state machine** [117]. In this view, the blockchain is not just a storage of information but a dynamic mechanism where transactions serve as inputs that result to transitions of the distributed ledger's state. Every succesful transaction creates new state that is formally accepted by the whole network. The usage of **formal modelling** is critical in this topic in order to give sufficient mathematical evidence for the security and stability of interoperability systems. Formal modelling removes ambiguity and ensures the correctness in comparing asset-bsed and data-based mechanisms under one theoretical framework [37] [39].

## 3.2 Semantic Modelling Using Labelled Transition Systems (LTS)

Labelled Transition System (LTS) is a fundamental mathematical structure used to model computations and dynamic behaviour of systems [105]. It represents systems as states and transitions labelled by actions.

However, in order to capture the complex behaviour of blockchain interoperability, the system is modelled as a **Probabilistic Lablled Transition System (pLTS)**. It is composed of quadruple

$$(S, Act, P, \rightarrow)$$

. In this framework, the transition relation ($\rightarrow$) does not only connects states but also refers to a probability distribution ($\Delta$) above the group of states ($S$). This probability distribution is crucial in modelling nondeterministic environments of consensus protocols and frame collisions [106].

The **observable behaviour** of the system is composed of the **traces**. They are the series of actions from the starting state [42]. This formalism allows the abstraction from the inside implementation to the outside communication [87]. These traces serve as the guidelines to analyse whether a protocol follows its intended semantic interoperability.

It is important to notice when two systems with differing structure representation or graph can share the same behaviour [12]. This concept is called **behavioural equivalence** wherein the states of different LTSs are treated as equal if they exhibit the same pattern of transitions such as what the theory of bisimulation describes [12] [42]. This abstraction allows the inquiry of comparison of protocols for asset-based and data-based interoperability in blockchains without limiting particular details of their internal implementation [87].

Nevertheless, the simple usage of state transition graphs can lead to the problem of state explosion that makes formal verification difficult [103]. For this reason, there is a need to a more expressive framework that allows algebraic manipulation and composition of processes.

## 3.3 Process Algebra for System Composition and Communication

**Process Algebra** serve as the main theoretical instrument of this research particularly the principles of communication, composition, and abstraction. The research will use it to formally analyse and manipulate operations of blockchains. This is illustrated as the study of concurrent systems or communicating systems inside an axiomatic and algebraic framework [18] [39]. Instead of just using visual diagrams, processes are embodies as formal *process terms* using a mathematical language that allows linear notation and mechanised manipulation [39].

### 3.3.1 Atomic Actions and Alphabet

System behaviour is constructed from a formally defined alphabet of atomic actions. Let:

$$D : \text{set of data}$$

$$P : \text{set of communication ports}$$

Atomic actions are defined as:

$$Send : s_p(d)$$

$$Read : r_p(d)$$

where $d \in D$ and $p \in P$ [104].

### 3.3.2 Communication and Handshaking

Communication between processes is governed by a communication function:

$$\gamma(s_p(d), r_p(d)) = c_p(d)$$

This formalises the handshaking paradigm where communication happens only through synchronisation between matching send and read actions. Isolated actions are disallowed to ensure correctness in system interaction [104].

### 3.3.3 Core Operators

In this framework, every system's process is composed of atomic actions (actions that has no duration) combined using the following operators: **Sequential Composition** ($x \cdot y$)**:** $x$ is perfomed first before $y$. **Alternative Composition** ($x + y$)**:** symbolises the nondeterministic choice between $x$ and $y$ operations. **Parallel Composition** ($x||y$)**:** symbolises the simultaneous execution of actions $x$ and $y$.

A fundamental feature of process algebra is its **compositionality** wherein a large system can be treated as combined smaller and simpler parts [39]. Using the **parallel operator** ( || ), individual blockchain network or protocol components can be modelled separately and then regouped to see the whole behaviour of the inter-operational system [39]. This modular technique allows the analysis of the connection between asset-based and data-based protocols as a series of concurrent processes [39].

### 3.3.4 Encapsulation and Abstraction

To control system visibility: **Encapsulation** ($\partial_H$) blocks unmatched internal actions. It is used in order to ensure these internal actions do not happen alone but as a part if a synchronized communciation that causes transition to the state of the whole composition. **Abstraction** ($\tau_I$) hides internal communications as silent steps ($\tau$) [39].

### 3.3.5 Timed and Reliable Behaviour

Since interoperability protocols operate in asynchronous environments, timing and reliability must be incorporated. The framework adopts the principles of **Positive Acknowledgement with Retransmission (PAR):** Successful delivery $\rightarrow$ acknowledgement ($ac$) Failure or delay $\rightarrow$ timeout ($to$) Recovery $\rightarrow$ retransmission To regulate behaviour, a priority operator ($\theta$) is introduced. It assigns lower priority to timeout actions than successful communications to prevent premature retransmissions.

Also, process algebra alone may not be sufficient in modelling interoperability mechanisms like HTLC. Because these mechanisms depend on timeouts, the formalism needs to be a Timed Process Algebra where in each process, there is a local variable *now* to keep track the passing of discrete time [106].

### 3.3.6 Operation Semantics

The meaning of process expressions is formally defined using **Structural Operational Semantics (SOS)**. Transition rules specify how processes evolve. This ensures that every behaviour is derivable and mathematically well-defined [104].

### 3.3.7 Behavioural Equivalence and Verification

In order to identify whether an implementation of a protocol matches to its expected behaviour, the concept of **behavioural equivalence** is used; particularly, the **branching bisimulation** ( $\leftrightarrow_{rb}$ ) [40] [39]. Branching simulation is needed because it maintains the branching structure of processes that is crucial in identifying deadlocks which can result to loss of value or message [40]. In order to prove this bisimulation between two different ledgers, **Cones and Foci methodology** must be used where every state in an implementation is mapped to a particular focus point at a specification that shows similar external behaviour [40].

## 3.4 Interoperability as System Composition

In process theory, interoperability can be formally described as a composition of two or more systems that work together to achieve a particular goal [57]. Under the framework of Process Algebra, this communication is symbolised by the parallel composition

$$(P||Q)$$

, wherein every blockchain network is treated as separate process that runs concurrently and independently from the other networks but with the ability to communicate to each other.

The communication inside this composition is run by the mechanism of **synchronisation** [76]. By means of a formal **communication function**, the individual actions of every blokchain gets to communicate. For example, the action that delivers the message (*send*) from a system needs to match the action of receiving (*read*) at the other system in order to complete a succefull communication.

The behaviour of the formed composition is treated as observable from the perspective of an outsider viewer [76]. Using the **abstraction operator** the complex details of the internal communication between blockchains are hidden or made as *silent steps* ($\tau$), so that what only remains are the external input-output relation of the whole inter-operational system [39]. The resulting traces

or series observable actions serve as the evidence if the operation of the composed systems remain consistent to the original semantic specification.

This framework of formal composition gives way to formally analyse and compare the two ways of blockchain communications: the asset-based and data-based interoperability.

## 3.5 Distinction Between Asset-Based and Data-Based Interoperability

In order to analyse the semantic behaviour of interoperability protocols, it is important to identify the difference between the two major category of interation in blockchain: asset-based and data-based interoperability.

Asset-based interoperability is formally illustrated as a mechanism to transfer ownership or value between different blockchain network. Under this model, a digital asset, like cryptocurrency, is formally burned or locked from its source chain in order to create an equal value at its destination chain. The primary goal here is the maintaining of the linear logic of the value wherein one asset cannot exit simultaneously in two different systems.

On the other hand, data-based interoperability is focused on the exhange of information or arbitrary data without needing to have a definite value. This type of interoperability allows a blockchain to read or verify the state of other blockchain, which is usually used to trigger **smart contract invocation**. Instead of transferring ownership, its focus is the delivery of messages from an external source towards an on-chain environment which is usually run by mechanisms like Oracle.

This difference leads to different behavioural implication for each protocol. In asset-based interoperability, the behaviour of the system is critcally tied to atomicity or the guarantee that the whole transaction between chains will be all successful or failure in order to avoid losing of value. In data-based interoperability, the behaviour is focused on semantic consistency and data integrity, wherein the goal is to make sure that the message received by the destination chain is authentic and true to its representation in the original state from where it came from.

The understanding of this difference will lead way to the creation of a formal model that will analyse their transitions.

# Chapter 4

# Methodology

The research methodology comprises a multi-stage pipeline. It will systematically process interoperability protocols from the technical specifications towards the formal mathematical verification. Every stage in this methodology is treated as sequential and dependent. Furthermore, the success of every following stage directly follows from the output of the previous stage. This ensures the logical connection of the whole process.

## 4.1 Formal Specification of Interoperability Protocols

The goal of this stage is to create a strict representation of representative protocols using algebraic language. This formalisation is important to remove details of their implementation that have no connection to the semantic behaviour. This allows comparison between asset-based and data-based mechanism.

### 4.1.1 Select Representative Protocols

Protocols that have wide adoptation and formal documentation will be used. For asset-based, HTLC; for data-based Oracle Relays; and for unified, IBC/XCM.

### 4.1.2 Identify Actions

Identify the elementary operations that makes up each protocol: **Asset-Based** ($A_{asset}$)**:** *lock* (locking of asset), *unlock* (moving asset out using secret), *refund* (returning of token after a timeout). **Data-based** ($A_{data}$)**:** *request* (asking of data), *verify* (confirmation of data integrity), *relay* (delivery of proven message to destination chain). Let:

$$D : \text{set of data (assets or messages)}$$

$$P : \text{set of ports (e.g., source chain, relayer, destination chain)}$$

Then:

$$A = s_p(d), r_p(d) | d \in D, p \in P$$

Protocol specific actions are mapped into this structure: Asset-based:

$$lock \rightarrow s_p(d)$$

$$unlock \rightarrow c_p(d) \text{ (succesful transfer)}$$

$$refund \rightarrow s_p(d) \text{ (reverse flow)}$$

Data-based:

$$request \rightarrow s_p(d)$$

$$verify \rightarrow \text{ internal communication}$$

$$relay \rightarrow c_p(d)$$

### 4.1.3 Encode Process

Form the logical protocol using the following operators: **Sequential ($\cdot$):** To signify strict sequential process. **Alternative ($+$):** To model branching of decision. **Parallel ($||$):** To model interaction between two different blockchain ledger. System components (e.g., sender, channel, receiver) are modelled as separate processes and composed as:

$$S||K||R||L$$

### 4.1.4 Encapsulation and Abstraction

To control system visibility: **Encapsulation set:**

$$H = s_p(d), r_p(d)|p \text{ is internal}$$

Ensures that unmatched actions are blocked. **Abstraction set:**

$$I = c_p(d)|p \text{ is internal}$$

Internal communications are hidden as silent steps ($\tau$).

### 4.1.5 Final System Representation

The complete system can be expressed as:

$$\tau_I(\partial_H(S||K||R||L))$$

This serves as the implementation model (IMP) that will be used for analysis.

## 4.2 Semantic Extraction Using Labelled Transition System

The goal of this stage is to give the process terms made from the preceding stage formal interpretation using Labelled Transition System (LTS). Through this, every algebraic term will have its corresponding dynamic model that shows all possible observable behaviour of the system.

### 4.2.1 Lay out Structural Operational Semantics (SOS)

SOS will be used in order to see the meaning of every operator in the process algebra using inference rules. Every rule is composed of premise and one conclusion:

$$\frac{premises}{conclusion}$$

### 4.2.2 Construct the LTS Model

For every specific protocol, LTS will be formed as a triple

$$(S, A, \rightarrow)$$

:

**States** ($S$)**:** set of states comprising the original process term and all terms that can be achieved from it (reachable states). **Labels** ($A$)**:** labels of transition that refers to atomic actions that the system executes. **Transition** ($\rightarrow$)**:** relation of transition ($s \xrightarrow{a} s'$) that shows change from one state to the another after the action ($a$). **pLTS** will be used when probabilisq behavious is included: captures message loss, corruption, and nondeterminism.

### 4.2.3 Extract Traces at Reachable States

**Reflexive States:** reflexive transitive closure ($\rightarrow *$) to identify all possible states that can be reached by the protocol from the starting state. **Traces:** get the finite sequences of actions

$$a_1, a_2, ..., a_n$$

that the system executed in order to illustrate the totally of its oberservable behaviour.

## 4.3 Description of Observable Behaviour

The goal of this stage is to formally identify the oberservable behaviour of the interoperability protocols using the extraction of traces and execution paths from the LTS mdoels. This survey will serve as the basis of comparison of asset-based and data-based mechanims at the noticeable level of their connecition.

### 4.3.1 Give meaning to Observable

The behaviour of a system is tread as oberservable if this comprises actions that can be seen or experience by an outside observer or adjacent system by means of communication. An observable behaviour is formally defined as an external specification (SPEC) that remains after applying the **abstraction operator ($\tau_I$)** in an implementation (IMP) [104]. In this methodology, the internal connections between chains are treated as **hidden actions** and are made as silent steps ($\tau$) in order to only see the connection of input and output between the system and its environenment . The rigour of the observation does not come from the absence of $\tau$ but on how abstraction lessens the complexity of the internal structure towards a simple SPEC that affirms the correctness of interoperability.

Internal calculations or local changes in the storage that cannot be seen at cross-chain communication will not be included in this survery.

### 4.3.2 Extract Traces and Execution Paths

**Traces:** Traces ($Tr(q)$) will be extracted by listing all possible series of oberservable actions that can be execute by the protocol from its origin state ($q_0$). A trace is a string $\sigma \in A*$ where there persist a sequential transitions towardsa new state. **Execution Path:** In order to survey the structure of the protocol deeper, execution paths or the total flow of the transition ($s \xrightarrow{a} s'$) that shows not only the series of actions but also points where there is nondeterministic choice or branch of decision.

**Justify as Basis of Interoperability:** The oberservable behaviour is the fundamental basis of interoperability because the communication between two blockchain is being treated as a black box interaction. In a chain's perspective, the internal implementation of the other chain has no corelation; what only matters is the actions being sent and being accepted by it in order to make sure the semantic consistency. In other words, the correctness of interoperability depends on whether the observable traces of the two system matches under the one specified protocol.

## 4.4 Evaluation Criteria

The goal of this stage is to set the formal guideline that will be used in comparing asset-bsed and data-based protocols. By means of identigying the level of similarity and difference of their behaviour, results of the analysation under one concept of semantic interoperability can be given interpretations.

### 4.4.1 Similarity, Difference, and Equivalence

**Equivalence:** Two protocol is equivalent if their process graph shows equal pattern of interaction to their environment. Formally, this is identified as an equivalence class ($|x| = y : x \sim y$) wherein the system belonging to this has one oberservable behaviour. **Similarity:** This refers to having some parts of the trace matching even if the structure of the process is not fully similar. **Difference:** This surfaces when a protocol has branching structure or deadlock behaviour that cannot be found in another even if they have matching strings of action.

### 4.4.2 Evaluation Criterias

**Trace Equivalence ( $=_{tr}$ ):** This criteria will be used to survey if the protocols can execute exactly the same series or strings of atomic action. Nonetheless, trace equivalence is considered as coarse because this does not capture the difference from deadlock or the timing of choices. **Bisimulation Equivalence ( $\sim$ ):** This the stricter criteria (finest). Under this criteria, two protocol need to have the same branching structure; meaning, in every step of a protocol, the other should respond to the same action and reach a state that is again equal.

### 4.4.3 Interpretation of the Results

Since asset-based and data-based protocols serve fundamentally different purposes (value transfer vs. message passing), this analysis *does not expect to find bisimulation* between them.

The expected contribution of this research is the *formal demonstration of semantic non-equivalence* between asset-based and data-based interoperability protocols, characterized by:

1. The presence of conservation invariants in asset-based traces that are absent in data-based traces
2. Strict causality requirements (lock before mint) in asset-based protocols vs. weak causality in data-based protocols
3. Atomicity failure semantics in asset-based vs. timeout/ACK semantics in data-based

The adherence to these formal criteria ensures that interoperability assessment does not depend only on technical connectivity but on mathematical correctness and semantic integrity of the system.

# Appendix A

# Use of Generative AI

The proponents utilised three generative AI: ChatGPT, Deepseek, NotbeookLM. However, there are no set rules on which to use for specific topics. The proponents use either of the three depending on their convenience.

## A.1 Brainstorming

Since generative AI tools like ChatGPT can quickly provide answers, it was used to ask trends that are happening in the field of Blockchain. It was asked to provide recent published papers that were checked by the proponents. It would give several research titles with the authors and links. The proponents would look at each of them to ensure their legitimacy. Extraction of the content of the provided researches were not used as there was no guarantee that the AI would give correct results. Tasks like those are prone to hallucinations.

## A.2 Searching

Another usage was the creation of keywords that can be used to search for papers from the online databases found in the internet. This made the curation of papers easier and faster. This, however, did not replace the manual reading of the existing papers especially their RRL section to find keywords, know more about the topic, and look for possible mentioned references.

## A.3 Formatting

During the writing of this paper, AI was used to automatically convert comma separated values into tables using latex syntax. This was done for convenience since the actual data was already created by the proponents.

Whenever a reference does not contain a DOI but contains its APA citation, ChatGPT is used to translate the APA format into bibtex format. This also applies to other citation formats to bibtex.

## A.4 Debugging

AI was also used in debugging error messages given by latex compiler during the compilation of the PDF format.

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